Henry Hsu

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SUMMARY

Industrial engineer (B.S. Industrial Engineering & Management, rank 6/64) who builds the systems it runs on. Modeled Job Shop Scheduling with genetic algorithms and Bayesian optimization, validated in Flexsim, and measured learned-scheduler collapse at scale (10% to 376% error). Cut response time 27.7% across 140+ APIs; scaled a pipeline to 20M+ records. M.C.S. at Rice, Dec. 2026.

EDUCATION

National Yang Ming Chiao Tung University (NYCU) - Top 3 Taiwan University **Sep. 2020 – Jun. 2024**

B.S., Industrial Engineering and Management (GPA: 4.07/4.30, Two-Time Dean's List) *Hsinchu, Taiwan*

Minor: Computer Science (Domain GPA: 4.13/4.30) | Class Rank: 6/64

- **IE Coursework:** Work Study (A), Production & Operation Management (A+), Facility Planning (A+), Quality Management (A+), Manufacturing Processes (A+), Engineering Economy (A+), Operations Research (A+)

Rice University - Top 20 U.S. University **Aug. 2025 – Dec. 2026 (Expected)**

M.C.S., Computer Science (GPA: 4.00/4.00) *Houston, TX*

WORK EXPERIENCE

Software Engineering Intern **Jun. 2026 – Aug. 2026**

Google *Taipei, Taiwan*

- **Test Infrastructure:** Built **3** integration test infrastructures from a **zero** baseline (**Java**, **Python**), onboarding **400+** test suites across **200+** smart home device types, with **90%+** unit test coverage for the infrastructure itself
- **Automated Failure Triage:** Built a service that triages scheduled test runs and posts root-cause summaries onto issue tickets, replacing manual log inspection and cutting failure narrow-down time by **20+ minutes** per issue; **shipped and adopted by the team**

Backend Engineer **Sep. 2024 – Aug. 2025**

MaiAgent Co., Ltd. *– Award-Winning B2B Software Startup* *Taipei, Taiwan*

- **Throughput & Latency:** Cut response time of 13 high-traffic endpoints by **27.7%** across a **140+** API surface by removing repeated work rather than adding capacity: SQL query refactoring, connection pooling, caching, and N+1 elimination
- **Capacity Scaling:** Scaled an indexing pipeline from **3M to 20M+** records at **3.5x faster** parsing, through a normalized schema and backward-compatible incremental refactors rather than a rewrite, so the system stayed live throughout
- **Defect Reduction:** Cut production hotfixes by **90%** initially and **50%** long term, building CI coverage from a **zero** baseline to **67%**
- **Requirements to Delivery:** Defined the product roadmap for a **10-person** company and owned features end to end, from feasibility evaluation through production release

PROJECTS

Job Shop Scheduling: Genetic Algorithm & Bayesian Optimization | *Python, Flexsim* **Apr. 2022 – Jun. 2022**

- **Scheduling Optimization:** Solved the Job Shop Scheduling Problem with a metaheuristic genetic algorithm in **Python**, tuning its hyperparameters by **Bayesian Optimization** rather than by hand
- **Simulation Cross-Validation:** Automated export of the optimizer output into a **Flexsim**-compatible format via Python file scripting, validating the schedules against the discrete-event simulation model instead of trusting the solver alone

Scheduling Models That Break at Scale: A Cross-Size Study | *PyTorch, Graph Neural Networks* **Mar. 2026 – Apr. 2026**

- **Failure Diagnosis:** Traced the collapse of learned schedulers on larger instances, relative scheduling error rising from **10%** in-distribution to **376%** out of it, to linear degree growth in the disjunctive graph
- **Architectural Fix:** Cut that error to a $\sim$**26%** ceiling, letting one 10×10-trained policy generalize to **7.5**$\times$ larger instances

LEADERSHIP

Student Association of Industrial Engineering and Management **Jun. 2022 – Jun. 2023**

President *NYCU*

- Ran four teams totaling **37** people and the association budget, delivering **20+** events in one year

SKILLS

Data & Optimization: Python, SQL, numpy, pandas, scikit-learn | Genetic Algorithms, Bayesian Optimization, Operations Research, Flexsim

Languages: Python, Java, C/C++, TypeScript | **Databases:** PostgreSQL, MongoDB, Elasticsearch | **Backend:** Django, FastAPI, Express.js

Testing: pytest, JUnit | **DevOps:** GitHub Actions, Docker, git, Shell Script | **ML/GenAI:** PyTorch, LlamaIndex, LangGraph